Lecture 13 - Ideal gas and the first law of thermodynamics

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In this lecture, we introduce the concept of the **internal energy** of a thermodynamic system. This is the sum of the kinetic energies of all its constituent particles, plus the sum of all the potential energies of interaction among these particles. We then have all the ingredients to present the **first law of thermodynamics**. We apply these ideas to the **ideal gas**.

A few slides in this chapter come from Chapter-19 of Young's textbook.
This PPT file will well define the scope of your tests.

1

The first law of thermodynamics

Recall the **principle of conservation of energy** for an isolated system:

$$\underbrace{\Delta KE + \Delta PE_{\text{gravity}} + \Delta PE_{\text{elastic}}}_{W} = W_{\text{done by nonconservative forces}}.$$

Now, **mechanical work** $W = \Delta KE + \Delta PE_{\text{gravity}} + \Delta PE_{\text{elastic}}$,

and $\boxed{W_{\text{done by nonconservative forces}} = -\Delta U}$ where U denotes the **internal energy** of a **thermodynamic system**.

E.g., U includes the total kinetic energy associated with the random translational motion of the molecules of an **ideal gas**. So,

$$W = -\Delta U.$$

Question.

Why is there a **minus sign** ?

The gain in mechanical work in the isolated system is done at the expense of the loss of its internal energy and vice versa!!

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The first law of thermodynamics (cont.)

Recall that we could also achieve similar results by only supplying **heat** Q to a **thermodynamic system**:

$Q = \Delta U$, i.e., we can use heat to increase the internal energy of a system.

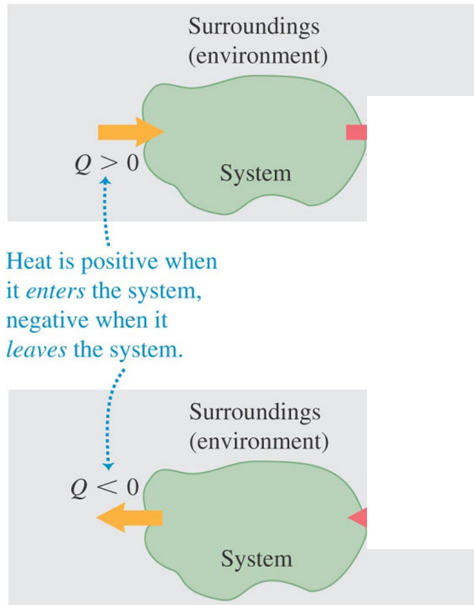
The **first law of thermodynamics** is a generalisation of the **principle of conservation of energy** to include energy transfer through **heat** Q into the system as well as **mechanical work** W, but the work W is done by the system. So the net change in the internal energy of the system is $\Delta U = Q - W$,

➔ $Q = \Delta U + W$.

work done by the system

In some textbook, it is written as $\Delta U = Q + W$.
➔ $Q = \Delta U - W$

work done on the system



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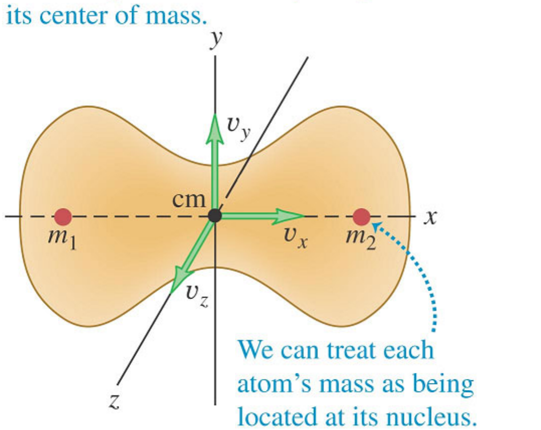
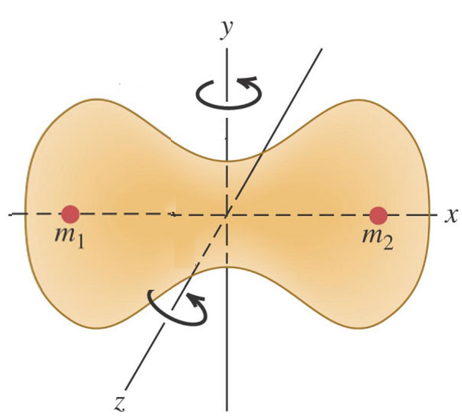
Internal energy of an ideal gas

Translational motion. The molecule moves as a whole; its velocity may be described as the x-, y-, and z-velocity components of its center of mass.



Rotational motion. The molecule rotates about its center of mass. This molecule has two independent axes of rotation. (Only 2 atoms in this figure)



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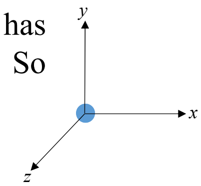
Internal energy of an ideal gas

If an **ideal gas** consists of only **monatomic molecules**, i.e., one-atom molecules, it is a **monatomic ideal gas**.

A **monatomic molecule** does not have any rotational motion. It has only kinetic energy associated with **random translational motion**. So the **internal energy** of n moles of a monatomic ideal gas,



Total number of particles in a sample

$$U = KE_{\text{translation}} = \frac{1}{2}Mv_{\text{rms}}^2 = \frac{3}{2}nRT.$$

Kinetic theory (previous chapter)

Since $N = nN_A$, we have $\boxed{U = \frac{3}{2}\left(\frac{N}{N_A}\right)RT = \frac{3}{2}N\left(\frac{R}{N_A}\right)T = N \times \frac{3}{2}kT.}$

Number of moles

6.022×10^{23}

Boltzmann constant

$$k = \frac{R}{N_A} = \frac{8.314}{6.022 \times 10^{23}} \approx 1.381 \times 10^{-23}\ \text{J/K}.$$

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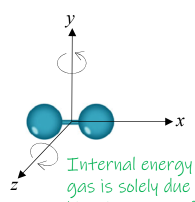
Internal energy of an ideal gas (cont.)

Kinetic Theory

If an **ideal gas** consists of only **diatomic molecules**, i.e., two-atom molecules, it is a **diatomic ideal gas**.

A **diatomic molecule** has kinetic energy associated with **random translational motion**, as well as **random rotational motion**. So the internal energy of a diatomic ideal gas with N diatomic molecules,

$$U = N \times \left(\overset{\text{translation}}{\frac{3}{2}kT} + \overset{\text{rotation}}{\frac{2}{2}kT}\right) = N \times \frac{5}{2}kT.$$



Internal energy of an ideal gas is solely due to the kinetic energy of the molecules in the container.

Since $N = nN_A$, the **internal energy** of n moles of a diatomic ideal gas,

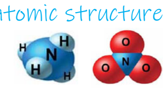
$$U = N \times \frac{5}{2}kT = (nN_A) \times \frac{5}{2}kT = \frac{5}{2}nRT.$$

$k = \frac{R}{N_A}$

$$n = \frac{M}{M_{\text{molar}}} = \frac{N}{N_A}\ \text{(mole)}.$$

mass in sample; molar mass; number of particles in sample; 6.022×10^{23}

(In this module we do not include polyatomic structure and vibrational motion.)



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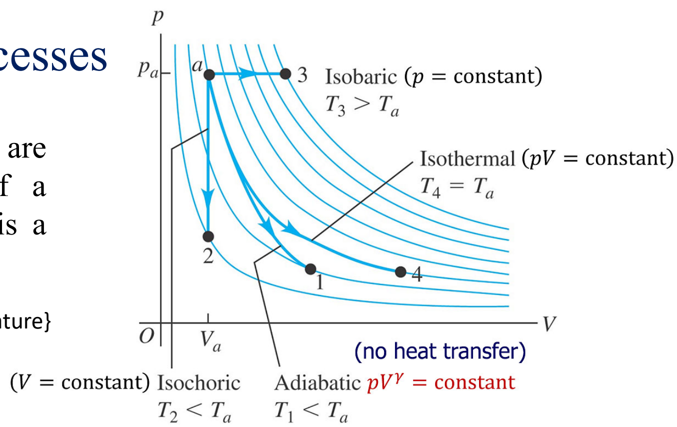
Thermodynamic processes

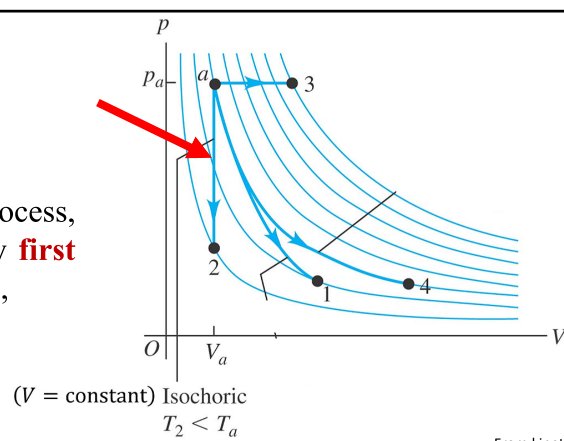


A process in which there are changes in the **state** of a **thermodynamic system** is a **thermodynamic process**.

State = { ?? }
State = {pressure, volume, temperature}

For an **ideal gas**, examples of **thermodynamic processes** include **isobaric** (constant pressure), **isochoric** (constant volume), **isothermal** (constant temperature), and **adiabatic** (no heat transfer) **processes**:

Isochoric process



$W = p \times \Delta V$

In a constant volume process, $dV = 0$ and $W = 0$. By **first law of thermodynamics**,

$\varDelta U = Q - W$ 0

➔ $\varDelta U = Q$.

Internal energy comes from the kinetic energy of the molecules in the container

For n moles of a **monatomic ideal gas**, $U = \frac{1}{2}Mv_{\text{rms}}^2 = \frac{3}{2}nRT$

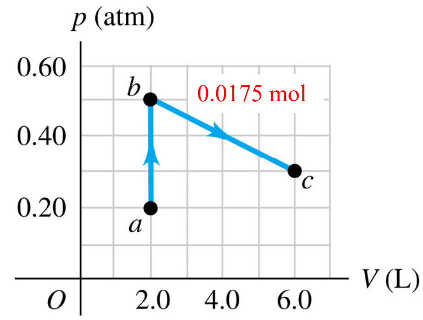
From kinetic theory (previous chapter)

$\varDelta U = Q = \cancel{n}C_V\cancel{\varDelta T}$ and also $\varDelta U = \frac{3}{2}\cancel{n}R\cancel{\varDelta T}$ ➔ $C_V = \frac{3}{2}R \approx 12.47\ \text{J mol}^{-1}\text{K}^{-1}$.

molar heat capacity at constant volume

1 Litre = 10cm x 10cm x 10cm
= 0.1m x 0.1m x 0.1m = 10^{-3} m^3

Problem. The process *abc* shown in the *pV*-diagram involve 0.0175 mol of an ideal gas. (i) What was the lowest temperature the gas reach in this process? Where did it occur? (ii) How much work was done by or on the gas from *a* to *b*? (iii) How much work was done by or on the gas from *b* to *c*? (iv) If 215 J of heat was put into the gas during the process *abc*, how many of those joules went into the internal energy?



Answer:

(i) From $pV = nRT$, $T_{\text{lowest}} = \dfrac{p_{\text{lowest}}V_{\text{lowest}}}{nR} = \dfrac{(0.2\times1.013\times10^5)(2\times10^{-3})}{0.0175\times8.314}$

$$= 278.50\ \text{K}\ (\text{at point } a)$$

(ii) $W_{ab} = 0$ because $(\Delta V)_{ab} = 0.$

(iii) $W_{bc} =$ area under the line *bc* (area of a trapezium)

$$= \left(\left(\frac{0.5\ \text{atm} + 0.3\ \text{atm}}{2}\right)\times1.013\times10^5\ \frac{\text{N}}{\text{m}^2}\right)\times(6\text{L}-2\text{L})\times10^{-3}\ \text{m}^3$$

$= 162.08$ J (work is done by the gas)

(iv) $\Delta U = Q - W$

$= 215 - 162.08 = 52.92$ J

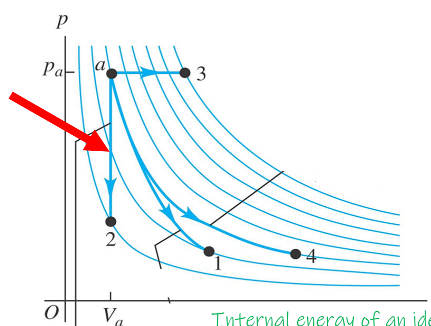
215 J of heat energy went into the gas, but only 52.92 J of energy stayed in the gas as increased internal energy, and 162.08 J left the gas as work done by the gas on its surroundings.

Isochoric process (cont.)

In a constant volume process, $dV = 0$ and $W = 0$. By **first law of thermodynamics**,

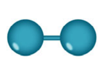
$$\Delta U = Q - \overset{0}{W}$$

➔ $\Delta U = Q.$



(V = constant) Isochoric
$T_2 < T_a$

Internal energy of an ideal gas is solely due to the kinetic energy of the molecules in the container.



For n moles of a **diatomic ideal gas**, $U = \dfrac{5}{2}nRT$. From kinetic theory

$$\Delta U = Q = \cancel{n}C_V\cancel{\Delta T} \text{ and also } \Delta U = \frac{5}{2}\cancel{n}R\cancel{\Delta T} ➔ C_V = \frac{5}{2}R \approx 20.79\ \text{J mol}^{-1}\text{K}^{-1}.$$

molar heat capacity at constant volume (isochoric)

Molar Heat Capacity at Constant Pressure

Molar Heat Capacity at Constant Volume

Relationship between C_p and C_V for an Ideal Gas

By the definition of the molar heat capacity at constant volume and its application, $\boxed{dQ = nC_V dT}$

When heat is given to the fixed-mass gas with the constrained volume, the pressure in the system will increase.

By the first law of thermodynamic: $\Delta U = Q - \cancel{W}^{\,0}$

➔ $\Delta U = Q.$

But the supplied heat $dQ = nC_V dT$ is given to the system.

So $dU = nC_V dT.$

(The changes in internal energy of an ideal gas is dependent on the temperature regardless of the type of process so $dU = nC_V dT$ can be used in other process later)

(The kinetic energy of n moles of substance is only dependent on its temperature.)

But for a constant-pressure process where $dV \neq 0$ the work dW done by the gas is not 0 because $dW = pdV$. By the ideal-gas equation of state $pV = nRT$, if p is kept as a constant, V will be directly proportional to T. So $p\,dV = nR\,dT$.

Apply the first law of thermodynamic to the constant-pressure process: $\Delta U = Q - W$

➔ $Q = \Delta U + W$

$W = F \times \Delta\ell = \left(\frac{F}{A}\right) \times (A \times \Delta\ell)$
$= P \times \Delta V$

$\cancel{n}C_p \cancel{dT} = nC_V dT + p\,dV$

$= \cancel{n}C_V \cancel{dT} + \cancel{n}R\,\cancel{dT}$

➔ $\boxed{C_p - C_V = R}$

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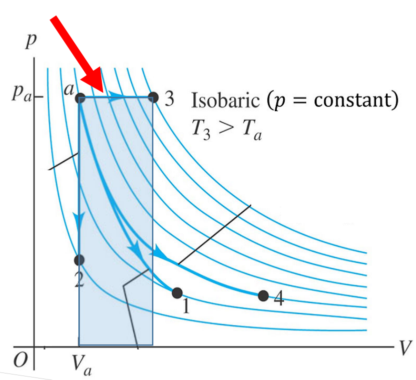
Isobaric process

In a constant pressure process, $W = p\Delta V$. By **first law of thermodynamics**,

$\Delta U = Q - W,$

➔ $Q = \Delta U + p\Delta V.$



For n moles of a **monatomic ideal gas**, $U = \frac{3}{2}nRT$ and $p\Delta V = nR\Delta T$.

So $Q = \frac{3}{2}\cancel{n}R\cancel{\Delta T} + \cancel{n}R\cancel{\Delta T}$, and also the supplied heat $Q = \cancel{n}C_p\cancel{\Delta T}$

➔ $C_p = \frac{5}{2}R \approx 20.79\ \text{J mol}^{-1}\text{K}^{-1}.$

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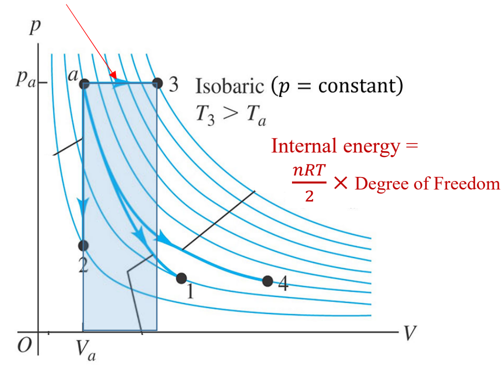
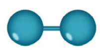
Isobaric process (cont.)



In a constant pressure process, $W = p\Delta V$. By **first law of thermodynamics**,

$$\Delta U = Q - W,$$

$$\rightarrow Q = \Delta U + p\Delta V.$$



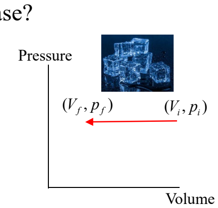
For n moles of a **diatomic ideal gas**, $U = \frac{5}{2}nRT$ and $p\Delta V = nR\Delta T$.

So $Q = \frac{5}{2}\cancel{n}R\cancel{\Delta T} + \cancel{n}R\cancel{\Delta T}$ and also $Q = \cancel{n}C_p\cancel{\Delta T}$

$$\rightarrow C_p = \frac{7}{2}R \approx 29.20 \text{ J mol}^{-1}\text{K}^{-1}.$$

Problem. A quantity of 2.00 mol of a monatomic ideal gas undergoes a compression during which the volume decreases from 0.0800 m^3 to 0.0500 m^3 while the pressure stays constant at a value of 1.8 x 10^4 Pa. (i) What is the work W? Is the work done by the gas or on the gas? (ii) What is the heat flow Q? Does heat enter or leave the gas? (iii) What is the internal energy change for the gas? Does the internal energy of the gas increase or decrease?

Answer:



For monatomic ideal gas at isobaric process, $C_p = \frac{5}{2}R$.

(i) $W = p(V_f - V_i)$ = (1.8 x 10^4 Pa) x (0.0500 m^3 – 0.0800 m^3) = –540 J

The minus sign tells us that work is done ***on*** the gas.

(ii) Supplied heat $Q = nC_p(\Delta T)$ … (1)

$pV = nRT \rightarrow \Delta(pV) = \Delta(nRT) \rightarrow p(\Delta V) + \underbrace{\cancel{V(\Delta p)}}_{0} = nR(\Delta T)$. So, $\Delta T = \frac{p\Delta V}{nR}$ … (2)

$$\Delta W = F \times \Delta\ell = \left(\frac{F}{A}\right)(A \times \Delta\ell) = P \times \Delta V$$

Sub (2) to (1), $Q = nC_p\left(\frac{p\Delta V}{nR}\right) = nC_p\left(\frac{W}{nR}\right) = \cancel{n}\left(\frac{5\cancel{R}}{2}\right)\left(\frac{W}{\cancel{nR}}\right) = \frac{5}{2}W = \frac{5}{2}(-540\text{ J}) = -1350\text{ J}$

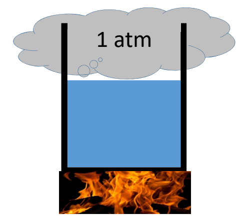
The minus sign tells us that heat **leaves** the gas.

(iii) $\Delta U = Q - W = (-1350\text{ J}) - (-540\text{ J}) = -810\text{ J}$. The internal energy *decreases*.

Or, $\Delta U = n\underbrace{C_V}_{\text{(monatomic)}}(\Delta T) = \cancel{n}\left(\frac{3\cancel{R}}{2}\right)\left(\frac{W}{\cancel{nR}}\right) = \frac{3}{2}W = \frac{3}{2}(-540\text{ J}) = -810\text{ J}$ (Same answer)

(This will have to be done by cooling.)

Problem. Water is heated in an open pen where the air pressure is 1 atm. The water remains as liquid, which expands by a small amount as it is heated. Determine the ratio of the work done by the water to the heat absorbed by the water. (For water, $c = 4186\ \text{J}/(\text{kg}\cdot\text{C}°)$, $\beta = 207\times10^{-6}\ (\text{C}°)^{-1}$ and $\rho = 1.00\times10^{3}\ \text{kg}/\text{m}^{3}$, and 1 atm = 1.013×10^5 Pa)



Answer:

Since the pan is open, the process takes place at constant (atmospheric) pressure 1 atm.

The work involved in an isobaric process is $W = p\Delta V$ … (1)

The heat absorbed by water is $Q = mc\Delta T$ … (2)

$$\frac{(1)}{(2)}:\ \frac{W}{Q} = \frac{p\Delta V}{mc\Delta T} = \frac{p(\beta V_i \Delta T)}{mc\Delta T} = \frac{p\beta}{\left(\frac{m}{V_i}\right)c} = \frac{p\beta}{\rho c} = \frac{(1.013\times10^{5})(207\times10^{-6})}{(1000)(4186)}$$

$$= 5.01\times10^{-6}$$

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Isothermal process

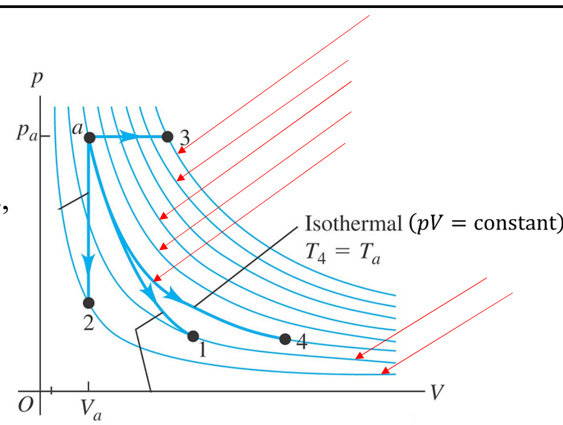
In a constant temperature process, $dT = 0$ so $\Delta U = 0$.

By **first law of thermodynamics**,

$$\cancel{\Delta U}^{\,0} = Q - W,$$

$$\rightarrow\ Q = W$$

$$pV = nRT$$
$$\rightarrow p = \frac{nRT}{V}$$



$$W = \int_{V_a}^{V_4} p\,dV = \int_{V_a}^{V_4} \frac{nRT_a}{V}\,dV = nRT_a \int_{V_a}^{V_4} \frac{1}{V}\,dV = nRT_a[\ln(V)]_{V_a}^{V_4}$$

Work done by the gas (system) on environment

$$= nRT_a(\ln(V_4) - \ln(V_a)) = nRT_a \ln\left(\frac{V_4}{V_a}\right)$$

Expansion. Gas is doing positive work on the environment

log 1 = 0
log 1.14 is positive
log 0.92 is negative

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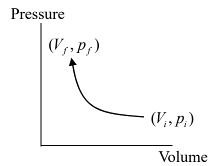
Problem. A 2.00-mol sample of helium gas initially at 300 K and 0.400 atm is compressed isothermally to 1.20 atm. Assume that the helium behave as an ideal gas. Find (i) the final volume of the gas, (ii) the work done on the gas, and (iii) the energy transferred by heat from the gas to the reservoir.

Answer:

(i) From $pV = nRT$, and T is a constant, $P_iV_i = nRT_i = P_fV_f$

$$V_f = \frac{nRT_i}{P_f} = \frac{2.00\text{ mol}(8.314\text{ J/K}\cdot\text{mol})(300\text{ K})}{P_f}$$

$$= \frac{4.988\times10^3\text{ J}}{1.2\times\left(1.013\times10^5\ \frac{\text{N}}{\text{m}^2}\right)} = 0.041\,0\text{ m}^3$$



(ii) The helium gas is being compressed so the work done on the gas is positive.

From $p_iV_i = p_fV_f$, $\frac{V_f}{V_i} = \frac{p_i}{p_f} = \frac{0.4}{1.2} = \frac{1}{3}$ ➔ $W = -nRT\ln\left(\frac{V_f}{V_i}\right) = -\left(4.988\times10^3\right)\ln\left(\frac{1}{3}\right) = 5.48\text{ kJ}$

Compression
Positive work is done by the environment on the gas.

(iii) For isothermal process, $\Delta U = 0$. So the work done on the gas must be discharged to the reservoir as heat.

The energy transferred by heat from the gas to the reservoir is 5.48 kJ.

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Adiabatic process (Supplied Heat $Q = 0$)

Differentiate $pV = nRT$ by product rule

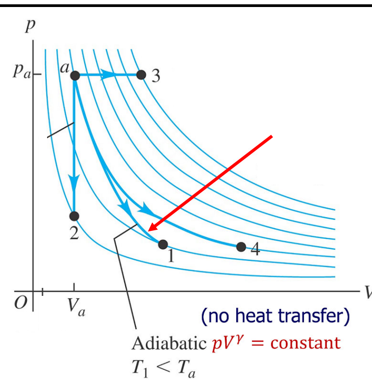
➔ $pdV + Vdp = nRdT$.

Recall $C_p - C_V = R$. So

$$pdV + Vdp = n(C_p - C_V)dT$$
$$= n\left(\frac{C_p - C_V}{C_V}\right)C_V dT$$
$$= \left(\frac{C_p}{C_V} - 1\right)nC_V dT \quad \dots (1)$$

By **first law of thermodynamics**,

$\Delta U = \overset{0}{Q} - W$ ➔ $W = -\Delta U$



That is, if there is no supplied heat Q, the work will be done at the expense of the internal energy of the gas. So

$W = -\Delta U$.

But $dU = nC_V dT$ regardless of the type of process.

Consider infinitesimal changes where $W = pdV$

➔ $pdV = W = -\Delta U = -nC_V dT$.

(1) becomes $pdV + Vdp = \left(\frac{C_p}{C_V} - 1\right)(-pdV)$

Internal energy of an ideal gas is only dependent on the kinetic energy with the total degree of freedom of the molecules inside the container

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Adiabatic process (cont.)

Let $\gamma = \frac{C_p}{C_V}$

$$pdV + Vdp = \left(\frac{C_p}{C_V} - 1\right)(-pdV)$$

$$pdV + Vdp = -(\gamma - 1)pdV$$

$$\rightarrow \cancel{pdV} + Vdp = -\gamma pdV + \cancel{pdV}$$

$$\rightarrow Vdp = -\gamma pdV$$

$$\rightarrow \frac{dp}{p} = -\gamma \frac{dV}{V}$$

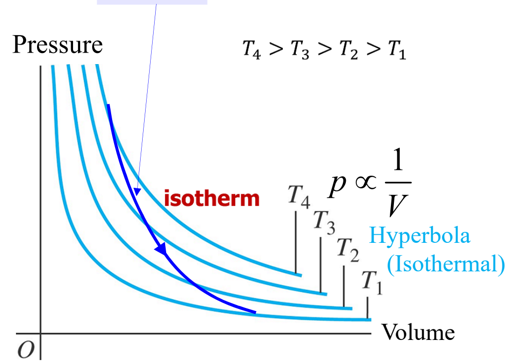
Integrating, $\displaystyle\int_{p_1}^{p_2} \frac{dp}{p} = -\gamma \int_{V_1}^{V_2} \frac{dV}{V}$

$$\ln\left(\frac{p_2}{p_1}\right) = -\gamma \ln\left(\frac{V_2}{V_1}\right) = \ln \frac{V_1^\gamma}{V_2^\gamma}$$

$$\rightarrow \boxed{p_2 V_2^\gamma = p_1 V_1^\gamma.}$$

Adiabatic

$p \propto \frac{1}{V^\gamma}$ $\quad \gamma > 1$



Why is there a drop in temperature when volume is increased adiabatically?

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Adiabatic process (cont.)

$$p_1 V_1^\gamma = p_2\, V_2^\gamma$$

Remarks

- With $p_1 V_1 = nRT_1$, and $p_2 V_2 = nRT_2$, we can derive (by algebra – next slide):

  $T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$, and, $p_1^{1-\gamma} T_1^\gamma = p_2^{1-\gamma} T_2^\gamma$.

In general, $C_p = C_V + R$.

- For a monatomic ideal gas, $C_V = \frac{3}{2}R$, $C_p = \frac{5}{2}R$, $\gamma = \frac{5}{3}$ for a adiabatic process.
- For a diatomic ideal gas, $C_V = \frac{5}{2}R$, $C_p = \frac{7}{2}R$, $\gamma = \frac{7}{5}$ for a adiabatic process.

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Adiabatic process

These are algebra!!

$$p_1V_1^\gamma = p_2V_2^\gamma \quad \ldots (1)$$

$$\frac{p_1V_1}{T_1} = \frac{p_2V_2}{T_2} \quad \ldots (2)$$

$$\frac{(1)}{(2)} \rightarrow \frac{V_1^{\gamma-1}}{\left(\frac{1}{T_1}\right)} = \frac{V_1^{\gamma-1}}{\left(\frac{1}{T_2}\right)}$$

$$\rightarrow \boxed{T_1V_1^{\gamma-1} = T_2V_2^{\gamma-1}}$$

$$p_1V_1^\gamma = p_2V_2^\gamma \quad \ldots (1)$$

$$(2)^\gamma \rightarrow \frac{p_1^\gamma V_1^\gamma}{T_1^\gamma} = \frac{p_2^\gamma V_2^\gamma}{T_2^\gamma} \quad \ldots (3)$$

$$\frac{(1)}{(3)} \rightarrow \frac{p_1^{1-\gamma}}{\left(\frac{1}{T_1^\gamma}\right)} = \frac{p_2^{1-\gamma}}{\left(\frac{1}{T_2^\gamma}\right)}$$

$$\rightarrow \boxed{p_1^{1-\gamma}T_1^\gamma = p_2^{1-\gamma}T_2^\gamma}$$

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Question. The largest bottle ever made by blowing glass has a volume of about 0.720 m^3. Imagine that this bottle is filled with air that behaves as an ideal diatomic gas. The bottle is held with its opening at the bottom and rapidly submerged into the ocean. No air escapes or mixes with the water. No energy is exchanged with the ocean by heat.

(i) If the final volume of the air is 0.240 m^3, by what factor does the internal energy of the air in the bottle increase?

Diatomic:
Degree of Freedom = 5

$$\gamma = \frac{C_p}{C_V} = \frac{\frac{5}{2}R + R}{\frac{5}{2}R} = \frac{7}{5} = 1.4$$

Answer:

No energy is exchanged with the ocean by heat → $Q = 0$ → adiabatic process.

$$P_iV_i^\gamma = P_fV_f^\gamma \rightarrow P_f = P_i\left(\frac{V_i}{V_f}\right)^\gamma \rightarrow P_f = P_i\left(\frac{0.720\text{ m}^3}{0.240\text{ m}^3}\right)^{1.40} = 4.66P_i$$

$$\frac{P_iV_i}{T_i} = \frac{P_fV_f}{T_f} \rightarrow T_f = T_i\frac{P_fV_f}{P_iV_i} = T_i\left(\frac{P_f}{P_i}\right)\left(\frac{V_f}{V_i}\right) = T_i\left(\frac{4.66P_i}{P_i}\right)\left(\frac{0.24}{0.72}\right) = T_i(4.66)\left(\frac{1}{3}\right) = 1.55T_i$$

Internal energy of the ideal gas is dependent on temperature, $U = nC_VT \rightarrow \frac{U_f}{U_i} = \frac{\cancel{nC_V}T_f}{\cancel{nC_V}T_i} = \boxed{1.55}$

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(ii) If the bottle is submerged so that the air temperature doubles, how much volume is occupied by air?

$V_i = 0.720\ \text{m}^3$

$\gamma = 1.4$

or $\boxed{T_1V_1^{\gamma-1} = T_2V_2^{\gamma-1}}$

Adiabatic Process

$\boxed{p_iV_i^{\gamma} = p_fV_f^{\gamma}}$

$\boxed{T_1V_1^{\gamma-1} = T_2V_2^{\gamma-1}}$

$\boxed{p_1^{1-\gamma}T_1^{\gamma} = p_2^{1-\gamma}T_2^{\gamma}}$

Answer:

Adiabatic process:

$$p_iV_i^{\gamma} = p_fV_f^{\gamma} \Rightarrow \frac{p_f}{p_i} = \left(\frac{V_i}{V_f}\right)^{\gamma} \quad \ldots (1)$$

$$nR = \frac{p_iV_i}{T_i} = \frac{p_fV_f}{T_f} \Rightarrow \frac{T_f}{T_i} = \frac{p_fV_f}{p_iV_i} = \left(\frac{p_f}{p_i}\right)\left(\frac{V_f}{V_i}\right) \quad \ldots (2)$$

$$= \left(\frac{V_i}{V_f}\right)^{\gamma}\frac{V_f}{V_i}$$

$$= \left(\frac{V_i}{V_f}\right)^{\gamma-1}$$

$$\frac{T_f}{T_i} = \left(\frac{V_i}{V_f}\right)^{\gamma-1}$$

$$2^{\frac{1}{0.4}} = \left(\frac{0.720\ \text{m}^3}{V_f}\right)^{1.4-1} = \left(\frac{0.720\ \text{m}^3}{V_f}\right)^{0.4\times\frac{1}{0.4}}$$

$$2^{\frac{1}{0.4}} = \frac{0.720\ \text{m}^3}{V_f} \Rightarrow \frac{0.720\ \text{m}^3}{V_f} = 2^{2.5} = 5.66$$

$$V_f = \frac{0.720\ \text{m}^3}{5.66} = \boxed{0.127\ \text{m}^3}$$

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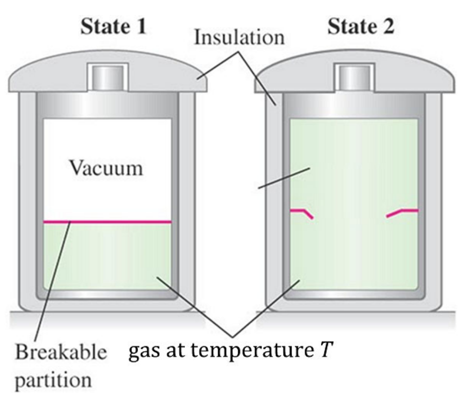
Free expansion

$W = \int p\,dV$ 0

Gas **expands freely** into the vacuum when the partition is broken. So, $W = 0$. And, due to the insulation, $Q = 0$.



By **first law of thermodynamics**,

$$\Delta U = Q - W$$ 0 0

$\Rightarrow \Delta U = 0.$ Therefore, the temperature of the gas stays constant, i.e., $\Delta T = 0$.

Question.

How does one represent a **free expansion** on the pV-diagram?

$pV = nRT$

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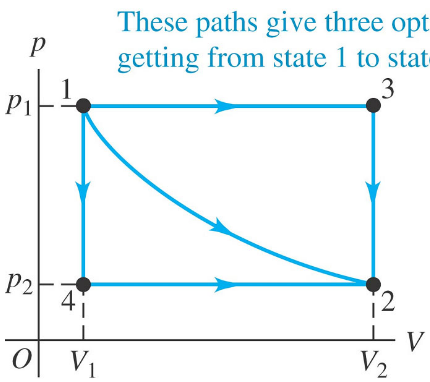
Paths between thermodynamic states

When a **thermodynamic system** changes from an **initial state** to a **final state**, it passes through a series of **intermediate states** – we call this series of states a **path**. When the **intermediate states** are all **equilibrium states**, the path can be plotted on a pV- diagram:

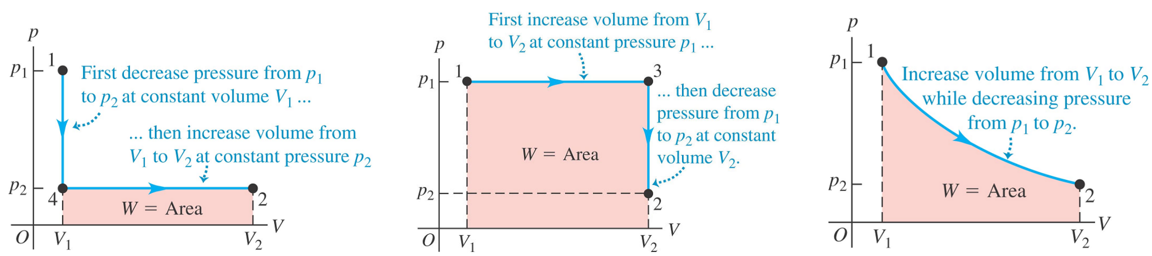


Remarks.

- **Reversible** (2 directions) versus **irreversible processes** (only 1 direction) .
- Given initial state 1 and final state 2, there are always infinitely many different possibilities for these intermediate states.

Paths between thermodynamic states (cont.)

The **work done** by the system depends not only on the **initial** and **final states**, but also on the **path**:



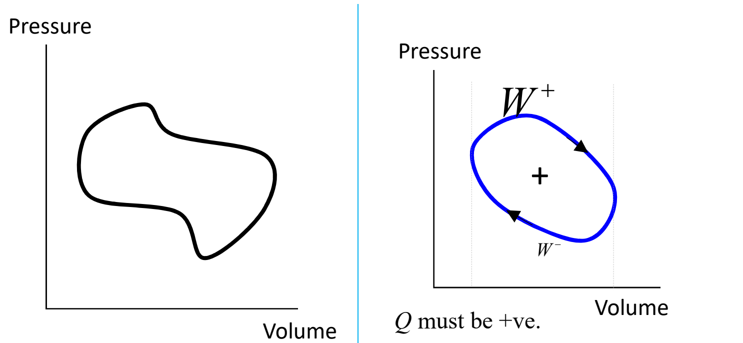
Like **work**, the **heat** added to a **thermodynamic system** when it undergoes a change of state also depends on the **path** from the **initial state** to the **final state**. Three examples are shown above.

Cyclic Process

The path on the PV curve is a closed loop.

Final state is same as initial state so there is no change in internal energy.

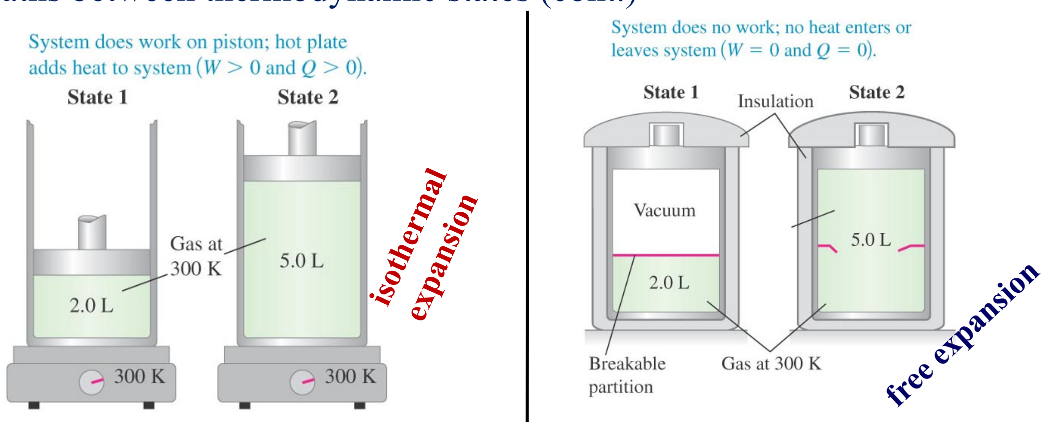
The sign of **net** Work W (net work done by the ideal gas) will be reversed if the direction of process is reversed, so the sign for the Heat Q will be reversed.



Q must be +ve.
Heat must enter the ideal gas to replenish the net work done by the ideal gas

Q must be –ve.
Heat must leave the ideal gas to drain away the excess energy due to the received work in the ideal gas

Paths between thermodynamic states (cont.)



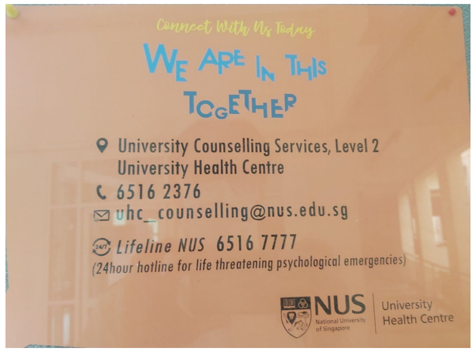
In general, while W and Q depend on the **path**, the **change in internal energy** ΔU of a system during any **thermodynamic process** depends only on the **initial** and **final states**, not on the **path** leading from one to the other – a **thermodynamic system** in a specific **state** has a unique **internal energy** U that depends only on that **state** (defined by p, V, T).

Helplines to de-stress

We are not superman/superlady. Sometime and somewhere in our life journey we will need help. Here they are:

Lifeline NUS (for life threatening psychological emergencies):
6516 7777 (24 hours)

6516 2376 (Office hours)
uhc_counselling@nus.edu.sg



Samaritans of Singapore Hotline: 1800 221 4444
Institute of Mental Health’s Helpline: 6389 2222
Singapore Association of Mental Health Helpline: 1800 283 7019